

TECHNICAL BULLETIN**DIXIE CHEMICAL**

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**EPOXY CURING AGENT 100
(ECA 100)****Typical Properties**

Properties	Value	Test Method
Molecular Weight (average)	162 - 166	
Specific Gravity @ 25°C, g/cc	1.205 ± 0.015	D-102 (ASTM D 4052)
Appearance	Clear to yellow liquid	D-104 (Visual)
Viscosity @ 25°C, cPs	50 - 80	D-150F
Color, Gardner	6 max.	D-150G
Volatiles, ppm	100 Max	D-160A
Total Anhydride, %	98.0 Min	D-160
Free Acid, %	2.0 max.	D-160B-2

Applications

ECA 100 is a light colored, low viscosity liquid epoxy curing agent designed for use in converting epoxy resins to highly crosslinked polymers with excellent physical and electrical properties. ECA 100 performs well when used in demanding applications, such as filament winding, pultrusion, and other forms of lamination, as well as electrical encapsulating, insulating, casting, and potting.

ECA 100 imparts ease of handling due to low viscosity, low toxicity and low volatility. It is a mixture, which is a liquid at normal room temperature. If the product crystallizes, it should return to liquid by heating 38-55°C (100-130°F) for 2-8 hours.

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ECA 100

For more information on the use of anhydrides like ECA 100 as epoxy curing agents, please consult the Technical Bulletin, "Formulating Anhydride-Cured Epoxy Systems", available from Dixie Chemical.

Formulation Example

ECA 100 was formulated with a variety of epoxy resins. The samples were formulated with 100 parts of epoxy resin, 84-117 parts ECA 100, and 1 part 1-Methylimidazole. Viscosity was measured at 25°C, and gel time was measured at 100°C. Samples were cured for 1 hour at 120°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes these results:

Epoxy type	ECA 100, phr	Viscosity @ 25 °C, cP	Gel Time ¹ , min	Tg ² , °C
Standard BPA Liquid Epoxy	85	965	41	140
Low Viscosity BPA Liquid	88	710	37	143
Cycloaliphatic Epoxy	117	144	42	221
Epoxy Phenol Novolac	93	990	36	143
Epoxy BPA Novolac	84	2830	34	153
BPF Liquid Epoxy	95	500	42	128

1. Shyodu Hot Pot (100g @ 100°C)

2. Cured 1 hr @ 120°C and 1 hr @ 220°C. Analyzed in DSC: 40°C/min to 250°C

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ECA 100**Electrical Performance Example**

When a formulated with 100 parts BPA liquid epoxy resin, 78-82 parts ECA 100, and 1 part accelerator and then cured for 1 hour at 95°C, and 3 hours at 150°C, the following properties are found:

Properties	Value
Dielectric Strength, volts per mil	480
Dielectric Constant, 120°C, 1 MHz	3.49
Power Factor, 120°C, 1 MHz	0.017
Volume Resistance, 25 °C, megohms-cms	3.9×10^{10}
Arc Resistance, seconds	70

Handling & Storage

ECA 100 will react with water to form diacids. This is normally undesirable, so ECA 100 should be stored in such a way that it is carefully protected from moisture contamination. When properly stored, ECA 100 has a shelf life of 24 months.

For more details on the design of bulk storage for ECA 100, consult the Dixie Chemical Company brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes.

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ECA 100**Health Hazards (cont.)**

In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

ECA 100 is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie Chemical Company representative for details.